

# Which Error Can the District Afford?

AP Statistics · Unit 3 · Topic 3.8 · B: 2026-12-08 · E: 2027-01-15 · TEACHER KEY

## CED TETHER

- **Skill 1.B** — Identify key and relevant information to answer a question or solve a problem.
- **Skill 3.A** — Determine relative frequencies, proportions, or probabilities using simulation or calculations.
- **Skill 4.A** — Make an appropriate claim or draw an appropriate conclusion.
- **Skill 4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- **EU UNC-5** — Probabilities of Type I and Type II errors influence inference.
- **LO UNC-5.A** — Identify Type I and Type II errors. [Skill 1.B]
- **EK UNC-5.A.1** — A Type I error occurs when the null hypothesis is true and is rejected (false positive).
- **EK UNC-5.A.2** — A Type II error occurs when the null hypothesis is false and is not rejected (false negative).
- Table of Errors:
  - $H_0$  true
  - $H_a$  true
  - Reject  $H_0$
  - Type I Error
  - Correct Decision
  - Fail to Reject  $H_0$
  - Correct Decision
  - Type II Error
- **LO UNC-5.B** — Calculate the probability of a Type I and Type II errors. [Skill 3.A]
- **EK UNC-5.B.1** — The significance level,  $\alpha$ , is the probability of making a Type I error, if the null hypothesis is true.
- **EK UNC-5.B.2** — The power of a test is the probability that a test will correctly reject a false null hypothesis.
- **EK UNC-5.B.3** — The probability of making a Type II error = 1 - power.
- **LO UNC-5.C** — Identify factors that affect the probability of errors in significance testing. [Skill 4.A]
- **EK UNC-5.C.1** — The probability of a Type II error decreases when any of the following occurs, provided the others do not change: i. Sample size(s) increases.
- ii. Significance level ( $\alpha$ ) of a test increases.
- iii. Standard error decreases.
- iv. True parameter value is farther from the null.
- **LO UNC-5.D** — Interpret Type I and Type II errors. [Skill 4.B]
- **EK UNC-5.D.1** — Whether a Type I or a Type II error is more consequential depends upon the situation.
- **EK UNC-5.D.2** — Since the significance level,  $\alpha$ , is the probability of a Type I error, the consequences of a Type I error influence decisions about a significance level.

**Essential question:** How should the consequences of false positives and false negatives shape a testing plan?

**DOK-3 skill:** 4.B · **FRQ pattern:** alpha-vs-power-which-error-tradeoff-fits-consequences

**DOK rationale:** Part (c) translates operating probabilities into contextual consequences, weighs two stakeholders' competing priorities, critiques two absolute claims, and proposes a design change that improves detection without erasing the tradeoff.

**Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in)** **DOK: 1**  
**→ 3** **Minutes: 45**

**Teacher does**

- Board slide up at the bell. Tally A, B, and need-more-information votes.
- Begin the video at minute 5; return at minute 33 to translate each probability into a decision and true state.
- Collect at the door. Score part (c) only, E/P/I.

**Students do**

- Choose a plan and name the prioritized error.
- Complete the follow-along, finish (a)–(c), revise, and turn in the sheet.

**Questions to ask**

- If  $p = 0.10$ , what decision creates the dangerous miss?
- What does the complement of 0.748 represent?
- What does Plan B trade per 1,000 tests?

**Adult role**

Solo teacher. Circulate 5 pairs. Require both consequences and both sides of the tradeoff.

**[WARN] WATCH FOR**

- Using  $1 - \alpha$  as the probability of a Type II error.
- Claiming that lowering alpha lowers both error probabilities when  $n$  is fixed.
- Choosing solely from the larger power without acknowledging its false-positive cost.

**SCORING PART (c) — E / P / I**

**Expected elements**

- **computes-both-error-risks** (required) — Uses  $\beta = 1 - \text{power}$  and Type I risk  $\alpha$
- **recommends-conditionally** (required) — Makes a stakeholder-linked, conditional recommendation
- **acknowledges-tradeoff** (required) — Contrasts 129 fewer misses with 40 more false replacements
- **critiques-both-claims** (required) — Explains why neither lower nor higher  $\alpha$  is cost-free
- **proposes-sample-size-change** (required) — Raises  $n$  to improve power at fixed  $\alpha$

|          |                                                                                                                                  |
|----------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>E</b> | Quantifies both errors, recommends conditionally, states the tradeoff and missing costs, critiques both claims, and raises $n$ . |
| <b>P</b> | Defends a plan but omits a consequence, critique, comparison, missing input, or design change.                                   |
| <b>I</b> | Uses $1 - \alpha$ for $\beta$ , calls either plan cost-free, or names a universal winner.                                        |

[STAR] TODAY'S PROBLEM — annotated

A district tests  $H_0 : p = 0.05$  versus  $H_a : p > 0.05$ , where  $p$  is the late-alert proportion. Both plans use  $n = 200$  and have the table's operating probabilities. Safety prioritizes avoiding retention when  $p = 0.10$ ; Budget prioritizes avoiding replacement when  $p = 0.05$ . No common cost scale exists.

**Dev:** "Use A. Smaller  $\alpha$  lowers both error risks."

**Lin:** "Use B. Its higher power has no cost when safety matters."

Test plans ( $n = 200$ )

| Plan | $\alpha$ | Power at $p = 0.10$ |
|------|----------|---------------------|
| A    | 0.010    | 0.748               |
| B    | 0.050    | 0.877               |

FIRST TAKE (5 min)

Which plan would you try first? Name the office you prioritized and one competing cost.

**Key:** Not graded. Either plan needs a conditional defense; neither is cost-free.

*Rules callout "ERROR PROBABILITIES TRADE OFF (keep on the page)" is printed on the student sheet.*

**DOK 1** (a) Describe a Type I error and a Type II error in this context. Identify what  $P(\text{reject } H_0 \mid p = 0.10)$  represents.

**Key:** A Type I error rejects  $H_0$  and concludes that more than 5 percent of alerts are late, causing the district to replace the system, when the true late rate is 0.05. A Type II error fails to reject  $H_0$  and retains the system when its true late rate exceeds 0.05; at the specified alternative, the true rate is 0.10. The conditional rejection probability at  $p = 0.10$  is the power against  $p = 0.10$ .

**DOK 2** (b) For each plan, calculate  $\beta$ . Then give the expected number of Type I errors per 1,000 repeated tests when  $p = 0.05$  and Type II errors per 1,000 repeated tests when  $p = 0.10$ .

**Key:** Plan A has  $\beta = 1 - 0.748 = 0.252$ ; per 1,000 repetitions it produces about  $1,000(0.010) = 10$  Type I errors when  $p = 0.05$  and  $1,000(0.252) = 252$  Type II errors when  $p = 0.10$ . Plan B has  $\beta = 1 - 0.877 = 0.123$ ; it produces about  $1,000(0.050) = 50$  Type I errors and  $1,000(0.123) = 123$  Type II errors in the corresponding states.

**DOK 3** (c) Adjudicate both claims. Explain why no plan is universally best; recommend one for a stated office using both error probabilities and their per-1,000 consequences. Name information needed for a district-wide choice and a change that improves power at fixed  $\alpha$ .

**Key:** No plan is universally best without consequence weights. For Safety, B is defensible: power is 0.877 versus 0.748, so Type II risk is 0.123 versus 0.252—123 versus 252 misses per 1,000 at  $p = 0.10$ . For Budget, A is defensible: Type I risk is 0.010 versus 0.050—10 versus 50 false replacements per 1,000 at  $p = 0.05$ . Thus B trades 129 fewer misses for 40 more false replacements. Dev ignores that lower  $\alpha$  raises Type II risk at fixed  $n$ ; Lin hides B's false-positive cost. The district needs relative costs and likely true states. Increasing  $n$  can raise power and lower  $\beta$  while holding  $\alpha$  fixed.

EXIT

One thing the video changed about my first take: \_\_\_\_\_